TAMP Project Overview

Purpose

The Truck Asset Matchmaking Platform is a role-based web application designed to connect freight owners with transporters through a centralized platform. It enables users to manage freight loads, truck assets, shipment requests, ratings, disputes, and verification processes while providing administrators with tools to oversee system operations. The application emphasizes a modular, user-friendly interface with role-specific functionality.

Scope

The system provides the following core capabilities:

- Secure user registration and authentication with role-based access control.
- Freight owner management of loads, shipment tracking, transport requests, and ratings.
- Transporter management of truck registrations, load requests, shipment updates, and ratings.
- Administrator management of users, disputes, verification requests, audit logs, and system settings.
- Local browser storage for data persistence through a service layer.
- Reusable layouts and UI components for a consistent user experience.

The current implementation is **frontend-focused**, with application data stored in **Local Storage** to simulate backend functionality.

Technology Stack

Technology	**Purpose**
React	Builds reusable and interactive user interfaces.
Vite	Provides fast development and build tooling.
React Router DOM	Manages navigation and protected routes.

Technology	Purpose
Tailwind CSS	Delivers responsive, utility-first styling.
ESLint	Maintains code quality and consistency.
React Icons	Provides scalable icons for the user interface.

Major Components

Authentication

- User Login
- User Registration
- Role-Based Access Control

Freight Owner

- Dashboard
- Load Management
- Match Requests
- Recommended Transporters
- Shipment Tracking
- Ratings

Transporter

- Dashboard
- Truck Registration
- Browse Available Loads
- Load Requests
- Shipments
- Ratings

Administrator

- Dashboard

- User Management
- Audit Logs
- Dispute Management
- Verification Management
- System Settings

Key Design Decisions

- **Role-Based Routing:** Restricts access to authorised pages based on user roles.
- **Local Storage Persistence:** Enables a self-contained application without requiring a backend server.
- **Service Layer Architecture:** Separates business logic from presentation, improving maintainability.
- **Reusable Components:** Promotes consistency and reduces code duplication across the application.
- **Modular Structure:** Organises features by user role, making the project easier to maintain and extend.
- **Centralised Routing:** Stores application routes in a single location for easier management.
- **Scalable Design:** Allows future integration with a backend database and API with minimal changes to the user interface.

Summary

TAMP is a modular, role-based logistics platform that demonstrates modern frontend development practices using React. Its architecture promotes maintainability, scalability, and code reuse while delivering dedicated workflows for freight owners, transporters, and administrators. The application's design provides a strong foundation for future integration with backend services, databases, and real-time logistics features.